First Terminal Examination - 2082

Class: XI | **Subject: Mathematics (0071)** | **Full Marks: 75**

Time: 3 hrs. | **SET 'B'** | **Pass Marks: 30**

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet. **[11 × 1 = 11]**

1. The inverse of the statement $q \to \sim p$ is

   a) $\sim q \to \sim p$ b) $\sim q \to q$

   c) $\sim p \to q$ d) $p \to \sim q$

2. For two simple statements r and s, the statement of De-Morgan's law for logic is

   a) $\sim (r \vee s) \equiv \sim r \wedge \sim s$ b) $\sim (r \wedge s) \equiv \sim r \vee \sim s$

   c) $\sim (r \to s) \equiv r \wedge \sim s$ d) Both (a) and (b)

3. If $A \subseteq B$ then $A \cap B =$

   a) B b) $\bar{A}$ c) A d) ϕ

4. For a non-negative real number a, $|x| \leq a$ implies

   a) $-a \leq x \leq a$ b) $-a < x < a$

   c) $-a > x > a$ d) $-a \geq x \geq a$

5. For $n \in \mathbb{Z}$, the value of $i^{4n+1} =$

   a) 1 b) i c) $-i$ d) -1

6. If $z = \frac{6 - 8i}{6 + 8i}$ then $z\bar{z} =$

   a) $\frac{6}{10} + \frac{8}{10}i$ b) $\frac{6}{10} - \frac{8}{10}i$ c) 1 d) 0

7. The quadratic equation whose roots are -1 and 2 is

a) $x^2 + x + 2 = 0$ b) $x^2 - 3x + 2 = 0$

c) $x^2 - x + 2 = 0$ d) $x^2 - x - 2 = 0$

8. The quadratic equation $px^2 + qx + r = 0$ has real and unequal roots if

a) $q^2 - 4pr < 0$ b) $q^2 - 4pr > 0$

c) $q^2 - 4pr = 0$ d) $q^2 > pr$

9. If $\lim_{x \to 2} \frac{x^n - 2^n}{x - 2} = 32$ for $n \epsilon z^+$, then $n =$

a) 4 b) 2 c) 3 d) 5

10. If $\lim_{x \to 0} \frac{\tan 7x}{\sin 5x} = \frac{1}{k}$ then $k =$

a) $\frac{7}{5}$ b) 1 c) $\frac{5}{7}$ d) 0

11. $\lim_{x \to 2} \frac{x - 2}{\ln(x-1)} =$

a) 2 b) -1 c) 0 d) 1

Group 'B'

Short Answer Questions. **[8 × 5 = 40]**

12. a) Define a contradiction. Show that $(p \wedge q) \wedge (p \wedge \sim q)$ is a contradiction. [1+2]

b) Write the negation of each of the following statements:

i) 4 is a prime number and $4 + 1 = 5$. [1]

ii) If an interval is closed then it includes both end points. [1]

13. a) For any two statements p and q, construct the truth table for (i) $p \Rightarrow q$ (ii) $\sim q \Rightarrow \sim p$. Can we say statements (i) and (ii) a logically equivalent? Give reason. [1+1+1]

b) For any two non-empty sets A and B, prove that:

$A \subseteq B \Longrightarrow \bar{B} \subseteq \bar{A}$ [2]

14. a) Define complement of a set in set builder form. If A and B are two non-empty subsets of universal set U, then prove that

$\overline{A \cap B} = \bar{A} \cup \bar{B}$. [1+2]

b) Find x and y if $(x+2)+(3-y)i = (-1+i)\cdot(3+2i)$. [2]

15. a) For any two complex numbers z and w, prove that:

$|z+w| \leq |z| + |w|$ [3]

b) Define absolute value of a complex number. Find the absolute value of $(3+4i)(4+i)$. [1+1]

16. a) For what value of 'k' will the equation $x^2+(k+2)x+2k=0$ have equal roots? [2]

b) If α and β are the roots of the equation of $px^2+qx+r=0,\ p \neq 0$, then prove that: (i) $\alpha+\beta = -\frac{q}{p}$ and (ii) $\alpha\beta = \frac{r}{p}$. [3]

17. a) If α and β are two roots of the equation $6x^2+7x+2=0$ then find $\frac{1}{\alpha}+\frac{1}{\beta}$. [1]

b) When does the limit of a function $f(x)$ at $x=a$ exist?

Evaluate: $\lim\limits_{x \to \theta} \frac{x\cos\theta - \theta\cos x}{x-\theta}$ [1+3]

18. a) Evaluate: $\lim\limits_{x \to a} \frac{\sqrt{x+a}-\sqrt{3x-a}}{\sqrt{x}-\sqrt{a}}$ [3]

b) If the function $f(x) = \begin{cases} \frac{\tan 8x}{x} & x \neq 0 \\ p & x = 0 \end{cases}$ is continuous at $x=0$, find the value of p. [2]

19. a) Test the continuity of the function $f(x) = \frac{|x-3|}{x-3}$ at $x=3$. If it is discontinuous, then mention the type of discontinuity. [2+1]

b) Find the points of discontinuity of $f(x) = \frac{x^2+4}{x^2-2x-8}$. [2]

Group 'C'

Long Answer Questions. **[3 × 8 = 24]**

20. a) Define difference of two non-empty sets in set builder form. For any three non-empty sets A, B and C, prove that:

$A - (B \cup C) = (A - B) \cap (A - C)$ [1+3]

b) If $A = [0, 6]$ and $B = (4, 8)$, find

i) $A \cap B$ ii) $A - B$ [2]

c) For $a, b \in \mathbb{R}$, prove that: $|a - b| \geq |a| - |b|$ [2]

21. a) For any complex number w, write the relation between w, $\overline{w}$ and $|w|$. If $w^2 = 7 - 24i$, then find the values of w in the form of $a + ib$. [1+3]

b) If p, q, r are rational, then prove that the roots of the equation $(p + r - q)x^2 + 2rx + (q + r - p) = 0$ are rational. [2]

c) If the roots of the quadratic equation $x^2 - 2lx + mn = 0$ be real and unequal, prove that the roots of $x^2 - 2(m + n)x + m^2 + n^2 + 2l^2 = 0$ will be imaginary. [2]

22. a) Evaluate: $\lim_{x \to 0} \frac{e^{mx} - e^{nx}}{x}$ [2]

b) Define continuity of a function $f(x)$ at $x = a$. Test the continuity of function

$$f(x) = \begin{cases} x^2 - 3 & for \quad x > 4 \\ 12 & for \quad x = 4 \\ 2x + 5 & for \quad x < 4 \end{cases} \quad \text{at } x = 4.$$

If it is not continuous, mention the type of discontinuity. If possible redefine $f(x)$ so as to make it continuous at $x = 4$. [1+2+1]

c) Evaluate: $\lim_{x \to \infty} (\sqrt{x - a} - \sqrt{x - b})$ [2]

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